

Chapter 5: Fejér Monotonicity and Fixed Point Iterations

Convex Analysis and Monotone Operator Theory

Based on Bauschke & Combettes (2017)

July 14, 2026

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Chapter Overview

- **Fejér Monotone Sequences:** Core concept for convergence analysis
- **Krasnosel'skii-Mann Iteration:** Convergence results for nonexpansive operators
- **Iterating Compositions:** Projection algorithms (POCS, Bregman, etc.)
- **Quasi-Fejér Monotonicity:** Relaxed conditions with error terms
- **Nonlinear Ergodic Theorems:** Cesàro averages and convergence

Key Insight: Fejér monotonicity provides a unifying framework for analyzing iterative algorithms without requiring strong convergence of the sequence itself.

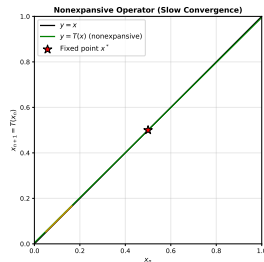
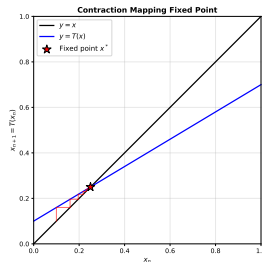
Motivation: Why Fixed Point Iterations?

Problem Setup:

- Find $x^* \in C$ where C is closed convex
- Solve $Tx = x$ where T is nonexpansive
- Minimize convex functions

Challenge:

- Direct solutions expensive/impossible
- Need iterative methods
- Understanding convergence critical



Definition: Fejér Monotone Sequences

Definition (Fejér Monotonicity)

Let C be a nonempty subset of $\mathcal{H}$ and $(x_n)_{n \in \mathbb{N}}$ a sequence in $\mathcal{H}$. The sequence $(x_n)_{n \in \mathbb{N}}$ is **Fejér monotone** with respect to C if

$$(\forall x \in C)(\forall n \in \mathbb{N}) \quad \|x_{n+1} - x\| \leq \|x_n - x\|.$$

Intuition:

- Distance to every point in C is decreasing
- Sequence approaches the set C uniformly
- Powerful convergence tool without explicit limit information

Example: Orthonormal sequence (e_n) in infinite-dimensional space is Fejér monotone w.r.t. $\{0\}$.

Key Properties of Fejér Monotone Sequences

Theorem (Proposition 5.4)

Let C be a nonempty subset of $\mathcal{H}$ and $(x_n)_{n \in \mathbb{N}}$ be Fejér monotone with respect to C . Then:

- ① $(x_n)_{n \in \mathbb{N}}$ is bounded.
- ② For every $x \in C$, $(\|x_n - x\|)_{n \in \mathbb{N}}$ converges.
- ③ $(d_C(x_n))_{n \in \mathbb{N}}$ is decreasing and converges.
- ④ $\|x_{n+m} - x_n\| \leq 2d_C(x_n)$ for all $m, n \in \mathbb{N}$.

Consequence: Distance function decreases monotonically, making strong convergence possible.

Weak Convergence Result

Theorem (Theorem 5.5)

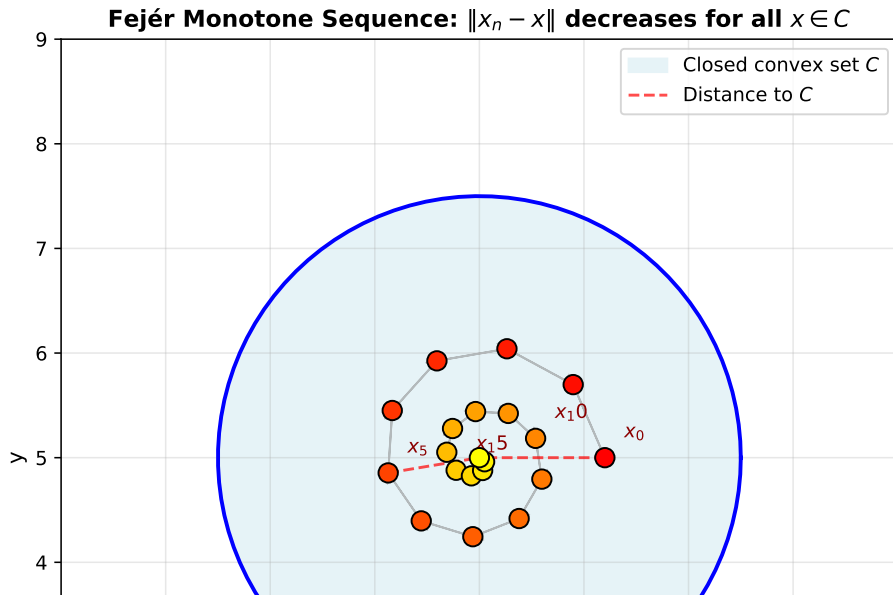
Let C be a nonempty subset of $\mathcal{H}$ and $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{H}$. Suppose that $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to C and every weak sequential cluster point of $(x_n)_{n \in \mathbb{N}}$ belongs to C . Then $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in C .

Proof Idea:

- Bounded sequence has weak cluster points (Banach-Alaoglu)
- All cluster points lie in C
- Decreasing distance to C forces unique cluster point

Fejér Monotonicity Visualization I

Fejér Monotonicity Visualization II



Strong Convergence Under Interior Condition

Theorem (Proposition 5.10 - Raik)

Let C be a subset of $\mathcal{H}$ with $\text{int } C \neq \emptyset$ and $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{H}$ that is Fejér monotone with respect to C . Then $(x_n)_{n \in \mathbb{N}}$ converges strongly to a point in $\mathcal{H}$ and

$$\sum_{n \in \mathbb{N}} \|x_{n+1} - x_n\| < +\infty.$$

Key Insight: When C has interior, Fejér monotonicity implies strong convergence!

Convergence to Projections

Corollary (Corollary 5.8)

Let C be a nonempty closed convex subset of $\mathcal{H}$, $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{H}$, and $x \in C$. Suppose that $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to C and $x_n \rightarrow x$. Then $P_C x_n \rightarrow x$.

Theorem (Theorem 5.12 - Linear Convergence)

Let C be a nonempty closed convex subset of $\mathcal{H}$, $(x_n)_{n \in \mathbb{N}}$ be Fejér monotone with respect to C , and $\kappa \in [0, 1[$ such that

$$(\forall n \in \mathbb{N}) \quad d_C(x_{n+1}) \leq \kappa d_C(x_n).$$

Then $(x_n)_{n \in \mathbb{N}}$ converges linearly to a point in C :

$$(\forall n \in \mathbb{N}) \quad \|x_n - x\| \leq 2\kappa^n d_C(x_0).$$

Affine Subspace Specialization

Theorem (Proposition 5.9)

Let C be a closed affine subspace of $\mathcal{H}$ and $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{H}$ that is Fejér monotone with respect to C . Then:

- ① $(\forall n \in \mathbb{N}) P_C x_n = P_C x_0$.
- ② *If every weak sequential cluster point of $(x_n)_{n \in \mathbb{N}}$ belongs to C , then $x_n \rightarrow P_C x_0$.*

Proof Sketch:

- Use Corollary 3.22(i) on affine subspaces
- Fejér monotonicity restricts to C
- Projection is constant along Fejér sequence

Firmly Quasinonexpansive Operators

Theorem (Proposition 5.13)

Let $(T_n)_{n \in \mathbb{N}}$ be a family of firmly quasinonexpansive operators from $\mathcal{H}$ to $\mathcal{H}$ with $C = \bigcap_{n \in \mathbb{N}} \text{Fix} T_n \neq \emptyset$, and $(\lambda_n)_{n \in \mathbb{N}}$ be a sequence in $]0, 2[$. Set $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n + \lambda_n(T_n x_n - x_n).$$

Then:

- ① $(\forall x \in C)(\forall n \in \mathbb{N}) \quad \|x_{n+1} - x\|^2 \leq \|x_n - x\|^2 - \lambda_n(2 - \lambda_n)\|T_n x_n - x_n\|^2.$
- ② $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to C .

Application: Projection operators onto closed convex sets are firmly nonexpansive!

The Krasnosel'skii-Mann Algorithm

Theorem (Theorem 5.14)

Let D be a nonempty closed convex subset of $\mathcal{H}$, let $T : D \rightarrow D$ be a nonexpansive operator with $\text{Fix}T \neq \emptyset$, and $x_0 \in D$. Set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = Tx_n$$

and suppose $x_n - Tx_n \rightarrow 0$. Then:

- ① $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in $\text{Fix}T$.
- ② If $D = -D$ and T is odd, then $(x_n)_{n \in \mathbb{N}}$ converges strongly to a point in $\text{Fix}T$.

Key Issue: Direct Banach-Picard iteration $x_{n+1} = Tx_n$ may not satisfy $x_n - Tx_n \rightarrow 0$ for nonexpansive T .

Groetsch's Theorem (Krasnosel'skii-Mann Relaxation)

Theorem (Theorem 5.15 - Groetsch)

Let D be a nonempty closed convex subset of $\mathcal{H}$, let $T : D \rightarrow D$ be a nonexpansive operator with $\text{Fix}T \neq \emptyset$, let $(\lambda_n)_{n \in \mathbb{N}}$ be a sequence in $[0, 1]$ such that $\sum_{n \in \mathbb{N}} \lambda_n(1 - \lambda_n) = +\infty$, and $x_0 \in D$. Set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n + \lambda_n(Tx_n - x_n).$$

Then:

- ① $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to $\text{Fix}T$.
- ② $(Tx_n - x_n)_{n \in \mathbb{N}}$ converges strongly to 0.
- ③ $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in $\text{Fix}T$.

Stepsize Condition: $\sum \lambda_n(1 - \lambda_n) = \infty$ ensures asymptotic regularity.

KM Iteration: Numerical Example

```
# Find fixed point via KM iteration
import numpy as np

def km_iteration(T, x0, lambda_n, N=100):
    x = x0
    history = [x0]
    residuals = []

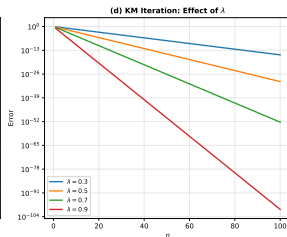
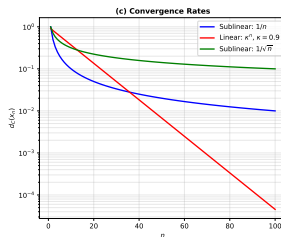
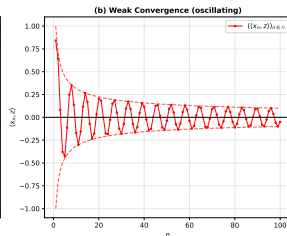
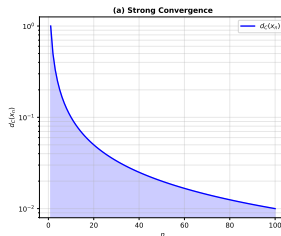
    for n in range(N):
        Tx = T(x)
        residuals.append(np.linalg.norm(x - Tx))

        x = x + lambda_n[n] * (Tx - x)
        history.append(x)

    return np.array(history), residuals

# Example:  $T(x) = 0.6x + 0.1$ 
T = lambda x: 0.6*x + 0.1
x0 = 2.0
lambda_n = [0.5] * 100

history, residuals = km_iteration(T, x0,
                                  lambda_n)
```



Averaged Operators

Definition

An operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is α -**averaged** if there exists a nonexpansive operator R and $\alpha \in]0, 1[$ such that

$$T = (1 - \alpha)\text{Id} + \alpha R.$$

Theorem (Proposition 5.16)

Let $\alpha \in]0, 1[$, let $T : \mathcal{H} \rightarrow \mathcal{H}$ be an α -averaged operator with $\text{Fix}T \neq \emptyset$, let $(\lambda_n)_{n \in \mathbb{N}}$ be a sequence in $[0, 1/\alpha]$ such that $\sum_{n \in \mathbb{N}} \lambda_n(1 - \alpha\lambda_n) = +\infty$, and $x_0 \in \mathcal{H}$. Then:

- ① $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to $\text{Fix}T$.
- ② $(Tx_n - x_n)_{n \in \mathbb{N}}$ converges strongly to 0.
- ③ $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in $\text{Fix}T$.

String-Averaged Relaxed Projections

Example (Example 5.21)

Let $(C_i)_{i \in I}$ be a finite family of closed convex sets with $C = \bigcap_{i \in I} C_i \neq \emptyset$. For every $i \in I$, let $\beta_i \in [0, 2[$ and set $T_i = (1 - \beta_i)\text{Id} + \beta_i P_{C_i}$. Set $x_0 \in \mathcal{H}$ and

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = \sum_{k=1}^p \omega_k T_{i(k,1)} \cdots T_{i(k,m_k)} x_n,$$

with weights $\omega_k > 0$ and $\sum_k \omega_k = 1$. Then $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in C .

Key Ideas:

- Combine projections onto multiple sets with relaxation
- String-averaged uses compositions
- Convex combinations of different orderings

POCS: Projections Onto Convex Sets

Theorem (Theorem 5.23)

Let D be a nonempty closed convex subset of $\mathcal{H}$, let m be a strictly positive integer, set $I = \{1, \dots, m\}$, let $(T_i)_{i \in I}$ be a family of nonexpansive operators from D to D with $\text{Fix}(T_1 \cdots T_m) \neq \emptyset$. Set $x_0 \in D$ and

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = T_1 \cdots T_m x_n.$$

Then $x_n - T_1 \cdots T_m x_n \rightarrow 0$ and there exist $y_1 \in \text{Fix} T_1 \cdots T_m$, etc., such that convergence relations hold (see text).

Application: POCS for finding intersection $C = \bigcap_i C_i$ via cyclic projections.

Corollary 5.24: POCS Algorithm

Corollary (POCS Algorithm)

Let m be strictly positive, set $I = \{1, \dots, m\}$, let $(C_i)_{i \in I}$ be a family of nonempty closed convex subsets of $\mathcal{H}$, let $(P_i)_{i \in I}$ denote their respective projectors, and $x_0 \in \mathcal{H}$. Suppose the intersection is nonempty and set

$$(\forall n \in \mathbb{N}) \quad x_{n+1} = P_1 \cdots P_m x_n.$$

Then the sequence converges weakly to the intersection.

Convergence: Weak convergence to intersection when it exists.

Bregman and Von Neumann-Halperin Algorithms

Corollary (Corollary 5.26 - Bregman)

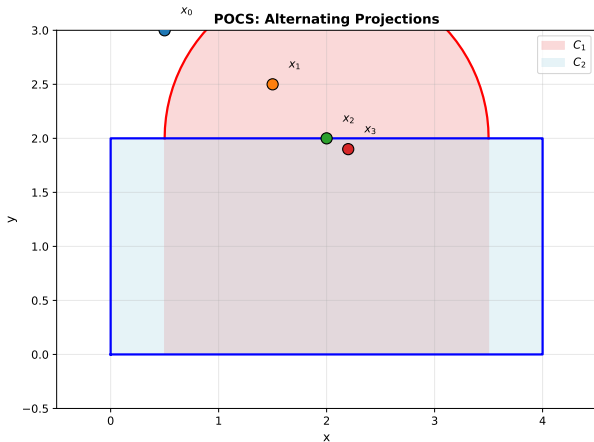
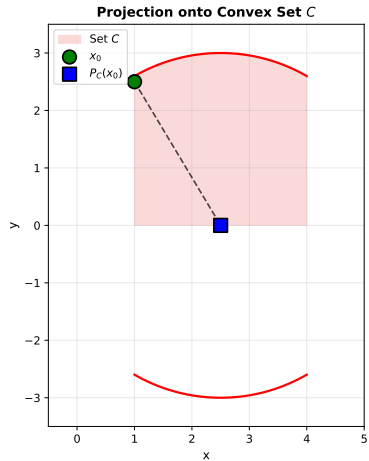
Let m be strictly positive, set $I = \{1, \dots, m\}$, let $(C_i)_{i \in I}$ be a family of closed convex subsets of $\mathcal{H}$ with $C = \bigcap_i C_i \neq \emptyset$. The cyclic projection sequence $x_{n+1} = P_1 \cdots P_m x_n$ converges weakly to a point in C .

Corollary (Corollary 5.30 - von Neumann-Halperin)

Let m be strictly positive, set $I = \{1, \dots, m\}$, let $(C_i)_{i \in I}$ be a family of closed affine subspaces of $\mathcal{H}$ with $C = \bigcap_i C_i \neq \emptyset$. Then $x_n \rightarrow P_C x_0$ (strong convergence).

Key Insight: For affine subspaces, we get strong convergence instead of weak!

Projection Algorithms Visualization I

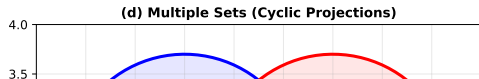
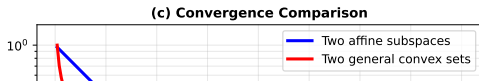
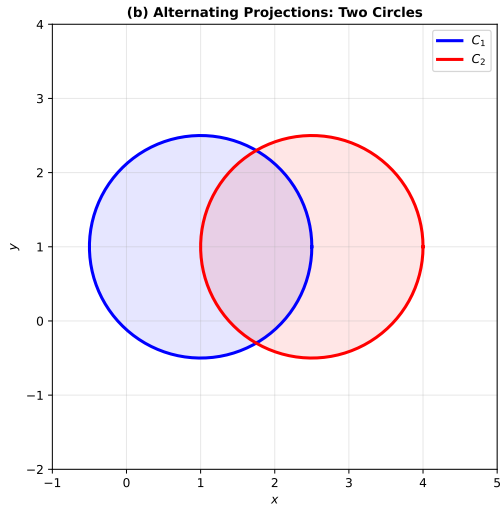
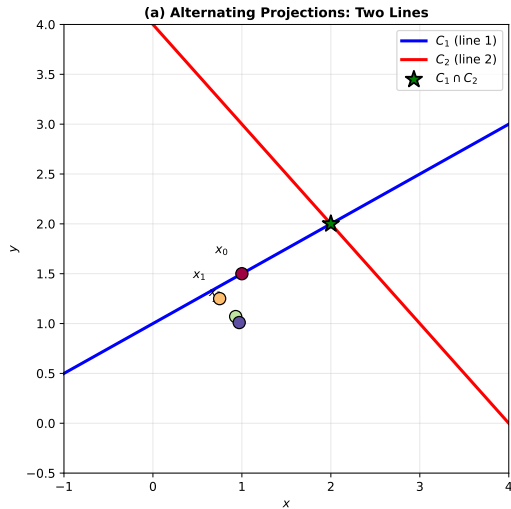


Projection Algorithms Visualization II

Left: Single projection onto a convex set. Right: Alternating projections (POCS) onto intersection of two sets.

Alternating Projections Convergence I

Alternating Projections Convergence II



Definition: Quasi-Fejér Monotonicity

Definition (Definition 5.32)

Let C be a nonempty subset of $\mathcal{H}$ and $(x_n)_{n \in \mathbb{N}}$ a sequence in $\mathcal{H}$. Then $(x_n)_{n \in \mathbb{N}}$ is **quasi-Fejér monotone** with respect to C if

$$(\forall x \in C)(\exists (\varepsilon_n)_{n \in \mathbb{N}} \in \ell^1_+(\mathbb{N}))(\forall n \in \mathbb{N}) \quad \|x_{n+1} - x\|^2 \leq \|x_n - x\|^2 + \varepsilon_n.$$

Key Difference: Allows small errors ε_n that sum to a finite value.

Theorem (Theorem 5.33)

Let $(x_n)_{n \in \mathbb{N}}$ be quasi-Fejér monotone with respect to C . Then:

- ① For every $x \in C$, $(\|x_n - x\|)_{n \in \mathbb{N}}$ converges.
- ② $(x_n)_{n \in \mathbb{N}}$ is bounded.
- ③ $(d_C(x_n))_{n \in \mathbb{N}}$ converges.
- ④ If every weak cluster point belongs to C , then $(x_n)_{n \in \mathbb{N}}$ converges weakly.

Quasi-Fejér Analysis with Perturbations

Theorem (Proposition 5.34)

Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be nonexpansive with $\text{Fix}T \neq \emptyset$, let $x_0 \in \mathcal{H}$, and let $(\lambda_n)_{n \in \mathbb{N}}$, $(e_n)_{n \in \mathbb{N}}$ satisfy $\sum_{n \in \mathbb{N}} \lambda_n(1 - \lambda_n) = +\infty$ and $\sum_{n \in \mathbb{N}} \lambda_n \|e_n\| < +\infty$. Set

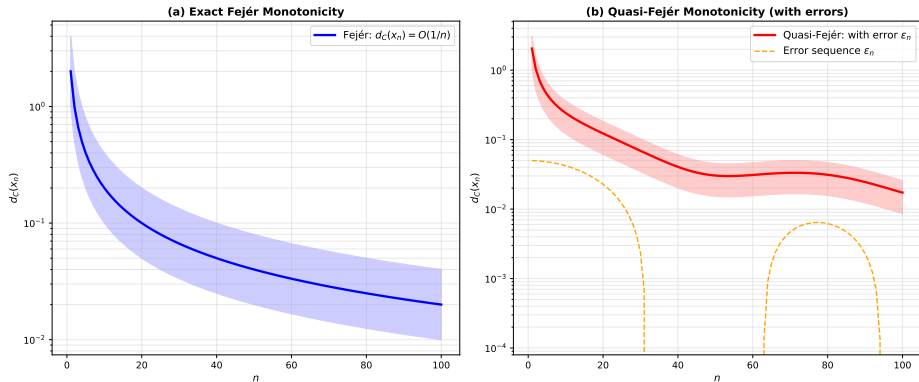
$$(\forall n \in \mathbb{N}) \quad x_{n+1} = x_n + \lambda_n(Tx_n + e_n - x_n).$$

Then:

- ① $(x_n)_{n \in \mathbb{N}}$ is quasi-Fejér monotone with respect to $\text{Fix}T$.
- ② $(Tx_n - x_n)_{n \in \mathbb{N}}$ converges strongly to 0.
- ③ $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in $\text{Fix}T$.

Robustness: Algorithm tolerates summable errors in the operator!

Quasi-Fejér Visualization I



Left: Exact Fejér monotonicity. Right: Quasi-Fejér with additive error sequences.

Cesàro Averages and Ergodic Theorems

Theorem (Theorem 5.36)

Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences in $\mathcal{H}$, let C be a nonempty subset, and set $C_m = \text{conv}\{y_k\}_{k \geq m}$. Suppose that:

- ① $(y_n)_{n \in \mathbb{N}}$ is quasi-Fejér monotone with respect to C .
- ② Distance from x_n to Cesàro hulls C_m goes to zero.
- ③ Every weak cluster point of $(x_n)_{n \in \mathbb{N}}$ is in C .

Then $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in C .

Key Idea: Combines quasi-Fejér monotonicity with Cesàro averaging.

Corollary 5.37: Cesàro-Averaged Sequences

Corollary (Corollary 5.37)

Let $(y_n)_{n \in \mathbb{N}}$ be quasi-Fejér monotone with respect to a nonempty subset C . For every $n \in \mathbb{N}$, let $(\alpha_{n,k})_{0 \leq k \leq n}$ be real numbers in $[0, 1]$ such that

$$(\forall n \in \mathbb{N}) \sum_{k=0}^n \alpha_{n,k} = 1 \quad \text{and} \quad (\forall k \in \mathbb{N}) \lim_{n \rightarrow \infty} \alpha_{n,k} = 0.$$

Set

$$(\forall n \in \mathbb{N}) \quad x_n = \sum_{k=0}^n \alpha_{n,k} y_k$$

and suppose every weak cluster point is in C . Then $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in C .

Application: Baillon's nonlinear ergodic theorem uses Cesàro average of iterates.

Baillon's Nonlinear Ergodic Theorem

Example (Example 5.38 - Baillon)

Let D be a nonempty closed convex subset of $\mathcal{H}$, let $T : D \rightarrow D$ be a nonexpansive operator with $\text{Fix } T \neq \emptyset$, and $x_0 \in D$. Set

$$(\forall n \in \mathbb{N}) \quad x_n = \frac{1}{n+1} \sum_{k=0}^n T^k x_0.$$

Then $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in $\text{Fix } T$.

Proof Sketch:

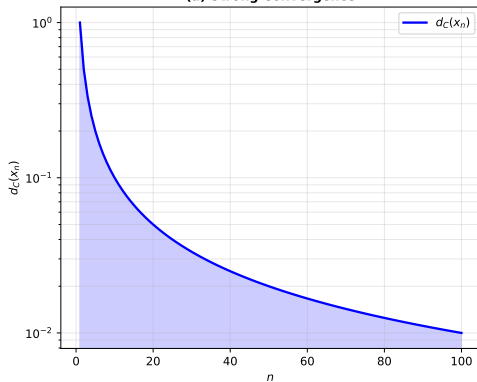
- 1 Set $C = \text{Fix } T$ and $y_k = T^k x_0$
- 2 Show (y_k) is Fejér monotone w.r.t. C
- 3 Apply Corollary 5.37 with $\alpha_{n,k} = 1/(n+1)$

Implication: Even when iterates diverge, their Cesàro average converges!

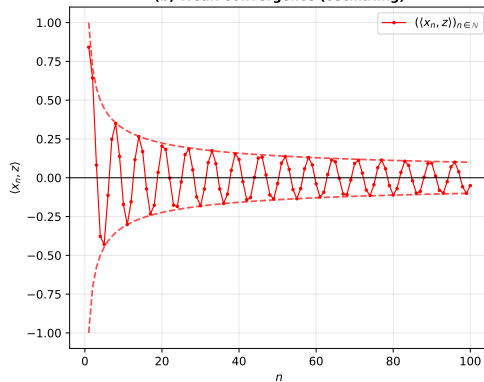
Convergence Rate Comparisons I

Convergence Rate Comparisons II

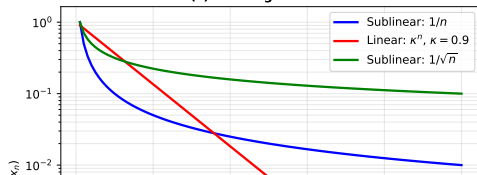
(a) Strong Convergence



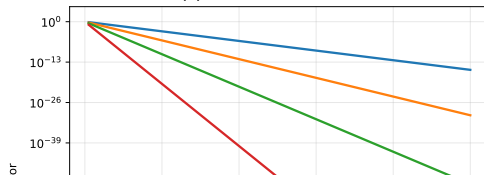
(b) Weak Convergence (oscillating)



(c) Convergence Rates



(d) KM Iteration: Effect of λ



Application 1: Finding Set Intersection

```
1 def pocs_algorithm(C1, C2, x0,
2                     max_iter=100, tol=1e-6):
3     """Projections Onto Convex Sets"""
4     x = x0
5     for n in range(max_iter):
6         # Project onto C1, then C2
7         x = project_onto_C1(x)
8         x = project_onto_C2(x)
9
10        # Check convergence
11        res = (linalg.norm(project_onto_C1(x)
12                           - x) +
13              linalg.norm(project_onto_C2(x)
14                           - x))
15        if res < tol:
16            break
17
18    return x
```

Example:

- $C_1 = \{x : \|x\| \leq 1\}$ (disk)
- $C_2 = \{x : x_1 + x_2 \leq 1\}$ (half-space)
- Find intersection

Advantages:

- Simple to implement
- Only needs projections
- Weak convergence guaranteed

Application 2: Solving Linear Systems

```
def solve_linear_system(A, b, x0,
                        max_iter=100):
    """Solve  $Ax=b$  via alternating projections"""
    x = x0
    for n in range(max_iter):
        # Project onto hyperplane  $Ax=b$ 
        res = A @ x - b
        x = x - (A.T @ res) / (
            linalg.norm(A, 'fro')**2)

    return x
```

Problem: Solve $Ax = b$

Reformulation:

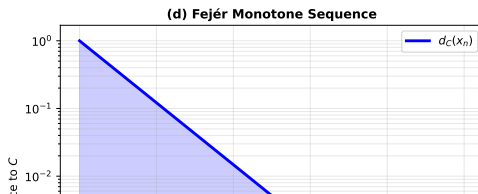
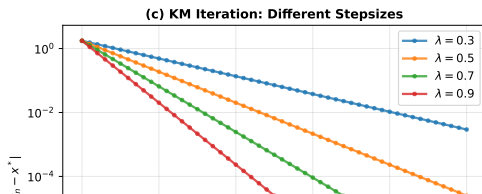
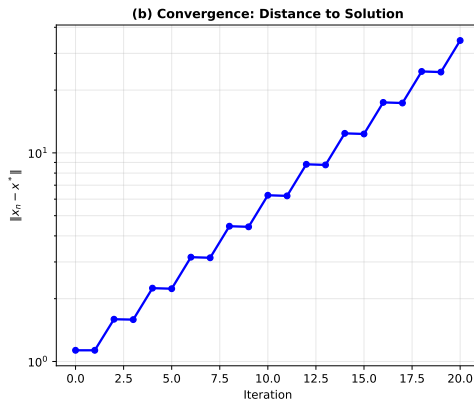
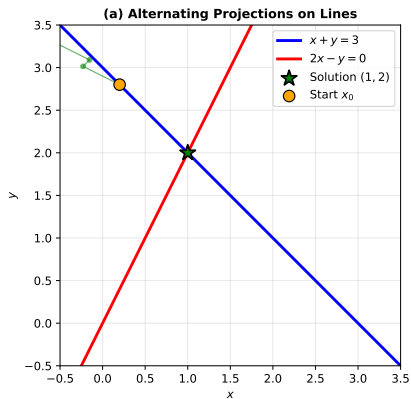
- $C_1 = \{x : Ax = b\}$ (hyperplane)
- $C_2 = \{x : \|x\| \leq R\}$ (ball)
- Iteratively project

Convergence:

- Weak to solution
- Works for least-squares

Numerical Example: Alternating Projections I

Numerical Example: Alternating Projections II



Practical Convergence Tips

- **Krasnosel'skii-Mann:** Use $\lambda_n = 1/2$ (simple and effective)
- **Stepsizes:** If unsure, use $\sum \lambda_n(1 - \lambda_n) = \infty$ (e.g., $\lambda_n = 1 - 1/(n+1)$)
- **Stopping Criterion:** Monitor distance to C or residual $\|Tx_n - x_n\|$
- **Multiple Projections:** POCS is robust but may converge slowly
- **Acceleration:** Use relaxation parameters $\beta_i \in [0, 2]$ for projections
- **Perturbations:** Quasi-Fejér allows summable errors (robust to noise)

Key Theorems Summary

- ➊ **Proposition 5.4:** Fejér sequences are bounded with converging distances
- ➋ **Theorem 5.5:** Weak convergence via cluster points in C
- ➌ **Theorem 5.14:** Banach-Picard with asymptotic regularity
- ➍ **Theorem 5.15 (Groetsch):** KM iteration with relaxation
- ➎ **Theorem 5.23:** POCS algorithm convergence
- ➏ **Corollary 5.24:** POCS for set intersection
- ➐ **Corollary 5.26 (Bregman):** Weak convergence for general sets
- ➑ **Corollary 5.30 (von Neumann-Halperin):** Strong convergence for affine subspaces
- ➒ **Theorem 5.33:** Quasi-Fejér with additive errors
- ➓ **Example 5.38 (Baillon):** Cesàro ergodic theorem

Conceptual Hierarchy

Fejér Monotonicity Chain:

- ① Sequence: distance decreases
- ② $\Rightarrow$ Bounded
- ③ $\Rightarrow$ Has weak cluster points
- ④ $\Rightarrow$ Weakly convergent

Algorithm Hierarchy:

- ① Nonexpansive T
- ② $\Rightarrow$ KM relaxation
- ③ $\Rightarrow$ POCS for projections
- ④ $\Rightarrow$ Convex feasibility

Central Theme: Fejér monotonicity is the common thread enabling convergence analysis of diverse iterative algorithms.

Chapter 5 Key Takeaways

- **Fejér Monotonicity:** Powerful convergence tool without explicit limits
- **KM Algorithm:** Solves the issue that direct Banach-Picard fails for nonexpansive operators
- **Projections:** POCS finds set intersections via simple alternating projections
- **Robustness:** Quasi-Fejér handles perturbations and rounding errors
- **Ergodic Averaging:** Cesàro means converge even when iterates don't

These tools form the foundation for optimization algorithms throughout applied mathematics.